COURSE DESCRIPTION FORM

INSTITUTION FAST School of Computing, National University of Computer and Emerging Sciences, Islamabad

Computer Science (BS) – **Spring 2024**

PROGRAM(S) TO BE EVALUATED

Course Description

Course Code	CS-2006
Course Title	Operating Systems
Credit Hours	3+1
Prerequisites by Course(s) and Topics	Data Structures
Grading Policy	**Absolute** grading
Policy about missed assessment items in the course	Retake of missed assessment items (other than midterm/ final exam) will not be held. For a missed midterm/ final exam, an exam retake/ pretake application along with necessary evidence are required to be submitted to the concerned committee. The examination assessment and retake committee decides the exam retake/ pretake cases.
Course Plagiarism Policy	Plagiarism in **project** or **midterm/ final** exam may result in F grade in the course. Plagiarism in an **assignment/quiz** will result in zero marks in **the whole** assignments/quiz category.
Assessment Instruments with Weights (homework, quizzes, midterms, final, programming assignments, lab work, etc.)	100% theory 3 credits and 1 credit separate lab Breakdown of Course Work (Total):

Assessment Item	**Number**	**Weight (%)**
Assignments	3-4	12
Sessional-1	1	16
Sessional-2	1	16
Quiz	4-6	8
Project	1	8
Final Exam	1	40

Course Instructors	Dr. M. Faisal Cheema, Dr. Adnan Tariq , Ms. Maryam Shahbaz, Ms. Rabail Zahid, Mr. Muhammad Aadil-ur-Rehman
Lab Instructors (if any)	Ms. Javeria Zia, Ms. Sana Razzaq, Ms. Hajira Uzair
Course Coordinator	Dr. M. Faisal Cheema, Ms. Maryam Shahbaz
URL (if any)	https://classroom.google.com/c/NjU2NjgyMDEwMzcw?cjc=njvi4pj
Current Catalog Description	Operating system is a course offered to BS CS. This course covers in detail many topics in Operating System design and implementation. Major topics to be covered are **Process Management, Thread Management, Process Synchronization, Deadlock**

	management, Memory Management and File Management etc. This is a programming intensive course and involves implementation and use of system calls, multithreaded applications, simulation of memory management, scheduling etc with emphasis on synchronization issues.
Textbook (or **Laboratory Manual** for Laboratory Courses)	**Operating System Concepts,** Author Silberschatz and published by John Wiley & Sons, 9th Edition
Reference Material	**Modern Operating System by Tenenbaum,** by Prentice Hall, (Latest Edition). **Operating System three easy pieces** by Remzi H. Arpaci-Dusseau **Operating Systems, 4th Edition,** William Stallings
Course Learning Outcomes	

A. Course Learning Outcomes (CLOs)

1. Describe, discuss and analyse, services provided by the modern Operating Systems.
2. Understand, design and implement solutions employing concepts of Processes and Threads.
3. Compare and contrast the commonly used for scheduling of tasks in operating systems and analyse different synchronization issues and implement synchronization mechanisms like Semaphores, Monitors, etc.
4. Understand and deploy OS concepts related to Virtualization.
5. Understand the operating system's resource management techniques, dead lock management techniques, memory management techniques.

B. Program Learning Outcomes

For each attribute below, indicate whether this attribute is covered in this course or not. Leave the cell blank if the enablement is little or non-existent.

1. Computing Knowledge:	Apply knowledge of mathematics, natural sciences, computing fundamentals, and a computing specialization to the solution of complex computing problems	✔
2. Problem Analysis:	Identify, formulate, research literature, and analyse complex computing problems, reaching substantiated conclusions using first principles of mathematics, natural sciences, and computing sciences.	✔
		✔

	3.Design/Develop Solutions:	Design solutions for complex computing problems and design systems, components, and processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.	
	4. Investigation & Experimentation:	Conduct investigation of complex computing problems using research-based knowledge and research-based methods.	✔
	5. Modern Tool Usage:	Create, select, and apply appropriate techniques, resources and modern computing tools, including prediction and modelling for complex computing problems.	✔
	6. Society Responsibility:	Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal, and cultural issues relevant to context of complex computing problems.	
	7. Environment & Sustainability	Understand and evaluate sustainability and impact of professional computing work in the solution of complex computing problems.	
	8. Ethics	Apply ethical principles and commit to professional ethics and responsibilities and norms of computing practice.	
	9.Individual and Teamwork:	Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings.	✔
	10.Communication	Communicate effectively on complex computing activities with the computing community and with society at large	
	11. Project Management and Finance	Demonstrate knowledge and understanding of management principles and economic decision making and apply these to one's own work as a member of a team.	

12. Lifelong Learning	Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological changes.	✔

C. Relation between CLOs and PLOs

		PLOs											
		1	2	3	4	5	6	7	8	9	10	11	12
CLOs	1	✔	✔		✔								
	2	✔	✔	✔	✔								
	3	✔	✔		✔	✔				✔			
	4	✔	✔	✔	✔	✔				✔			
	5	✔	✔	✔	✔	✔				✔			✔

Topics Covered in the Course, with Number of Lectures on Each Topic (assume 15-week instruction and one-hour lectures)

Topics to be covered:

List of Topics	No. of Weeks	Contact Hours	CLO(s)
Introduction, Overview OS Services, Docker Containers	**2**	**3**	**1.2**
Processes	**2**	**6**	**1**
IPC, Pipes	**1**	**3**	**1**
CPU Scheduling	**1**	**3**	**1,2**
Threads, Deadlocks	**2**	**6**	**2**
Synchronization, Bakery Algorithms	**1**	**3**	**2**
Semaphores	**1**	**3**	**1,2**

	Memory Management & its Data Structure	**1**	**3**	**1,2**
	Paging & Segmentation	**1**	**3**	**1,2**
	File System Interface & Implementation	**1**	**3**	**1,2**
	Advanced Topics	**1**	**3**	**3**
	Project Demo	**1**	**3**	**2,3**
	Total	**15**	**45**	

Laboratory Projects/Experiments Done in the Course	Introduction of lab and shell scripting, introduce the concept of processes, process creation and handling, hands on practice of exec system call, communication techniques between processes, introduction to signal and signal handling, concept of threading, how modern operating systems utilizing threading for parallel processing, how to synchronize processes and threads. Evaluation is managed with problem approach evaluated by lab work, practical quizzes and lab project.			
Programming Assignments Done in the Course	Yes, concepts related to Threads/Processes/Scheduling implemented using a preferred programming language selected by student (e.g. C/C++)			
Class Time Spent (in hours)	**Theory**	**Problem Analysis**	**Solution Design**	**Social and Ethical Issues**
	28	10	5	2
Oral and Written Communications	Every student is required to submit at least __1__ written reports of typically __8_ pages and to make __1___ oral presentations/demo of typically __15___ minute's duration. Include only material that is graded for grammar, spelling, style, and so forth, as well as for technical content, completeness, and accuracy.			

Lab/ Practical Component of the course

LAB CONTENTS (Lab/ Practical):

Weeks	Contents/Topics	**Courseware Events (MM/ IT Lab/Case Study/ Assignment/ Presentation etc.)	Comments (if any)
Week-01	Shell Scripting	Lab Exercise # 01	
Week-02	Dockers	Lab Exercise # 02	
Week-03	Process Creation – mainly fork(), wait(), exit() system calls	Lab Exercise # 03	
Week-04	Exec() System call – a family of six functions	Lab Exercise # 04	
Week-05	Inter-process communication	Lab Exercise # 05	
Week-06	Named pipes, Dup System calls	Lab Exercise # 06	
Week-07	Introduction to Threads	Lab Exercise # 07	
Week-08	Multithreading	Lab Exercise # 08	
Week-09	----------	Mid Exams	
Week-10	Thread Attributes	Lab Exercise # 09	

Week-11	Mutex	Lab Exercise # 10	
Week -12	Detailed Exercise on Mutex	Lab Exercise # 11	
Week 13	Semaphores	Lab Exercise # 12	
Week-14	Detailed Exercise on Semaphore	Lab Exercise # 13	
Week-15	Project		

Practical/ Programming Work/ Tools: Linux (Ubuntu), C/C++, gcc/g++ packages